

FairFlow Guardian

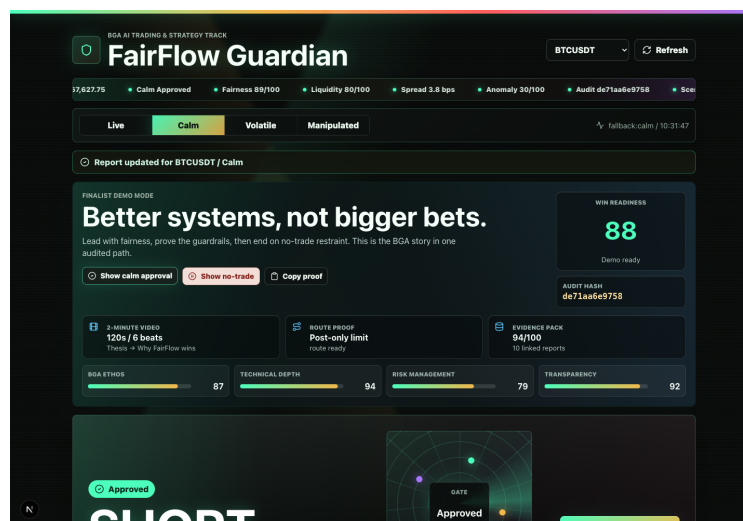
Complete hackathon walkthrough for the BGA AI Trading & Strategy Track

This PDF explains what the project does, how the app should be demoed, and how the underlying agents, fairness checks, risk controls, competition runway, route router, evidence pack, and audit proofs fit together.

Hackathon Readiness Runbook: a judge-facing guide mapped to BGA ethos, technical depth, strategy design, risk management, transparency, and verifiability.

READINESS	CURRENT AUDIT	ROUTE
87.8/100	7193e89a69c5	Post-only limit
demo ready	approved / SHORT	route ready

FairFlow Guardian is not trying to win a pure PnL contest. It is an explainable, paper-only trading safety layer that uses public market data, AI-style agents, retail fairness checks, route fairness, adversarial probes, and audit hashes to prove when a trade is fair enough to review and when the healthiest market action is no trade.



1. Executive Summary

FairFlow Guardian is a trading safety layer, not a profit-maximizing black box.

- No-trade is treated as a successful protective outcome.
- Execution is paper-only and cannot bypass the Fairness Passport, Policy Studio, route router, or audit gate.
- Every major dashboard claim has a machine-readable proof endpoint tied to the same audit hash.
- The system documents model limits and validation artifacts instead of overstating predictive accuracy.

Current calm audit:

FINAL ACTION	FAIRNESS	HIDDEN COST
SHORT	89/100	3.8 bps
approved	fair_to_execute	spread plus impact pressure

2. Hackathon Readiness Runbook

Use this sequence when a judge asks what the project does.

Criterion	Score	What to show	Proof
Bga Ethos	86/100 - ready	Start here to show the project reduces information asymmetry instead of chasing opaque PnL.	/api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/receipt, /api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/retail-cohorts
Technical Depth	94/100 - ready	Use this to defend that the project is more than a thin API wrapper.	/api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/model-provenance, /api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/anchor-proof
Risk Management	79/100 - watch	Switch to manipulated conditions to prove blocked execution is a successful safety outcome.	/api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/policy-stress, /api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/execution-router
Transparency	92/100 - ready	Copy any proof endpoint to show the UI narrative is backed by machine-readable artifacts.	/api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b, /api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/evidence-pack

Start from the Competition Runway panel in the dashboard: it collects the BGA thesis, proof-copy action, readiness score, evidence pack status, route proof, and calm/no-trade scenario controls in one judge-friendly surface.

Recommended opening line

FairFlow approved only a guarded paper short for BTCUSDT, then proves why that route is fair enough for retail users.

The detailed runbook spans the next pages. Every step has a UI action, expected result, underlying mechanism, and proof URL.

3.1 Step-by-Step Runbook

Follow these actions in the dashboard and explain the mechanism under each surface.

Step	Action	Underlying mechanism	Proof
1. Open with the thesis	Open Judge Mode Brief and read the one-sentence pitch.	The brief is generated from the same audit hash, impact ledger, fairness receipt, cohort simulator, and anchor proof.	/api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/judge-brief
2. Show the audited decision	Use the top decision panel, fairness score, hidden cost, and audit hash.	Market data is transformed into spread, depth, volatility, RSI, funding, impact, and liquidity metrics before the agents vote.	/api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b
3. Prove retail fairness	Open Fairness Receipt and Retail Cohort Simulator.	The app checks information parity, execution parity, manipulation exposure, risk protection, auditability, hidden cost, account size, and margin pressure.	/api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/retail-cohorts
4. Show the model and data card	Open Model and Data Provenance.	The card separates live/fallback data, derived metrics, ML signals, risk models, execution guards, and audit infrastructure.	/api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/model-provenance

3.2 Step-by-Step Runbook

Follow these actions in the dashboard and explain the mechanism under each surface.

Step	Action	Underlying mechanism	Proof
5. Explain fair route selection	Open Fair Execution Router.	Route candidates are scored for slippage, fill probability, information leakage, manipulation exposure, and retail route fairness.	/api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/execution-router
6. Stress the governance policy	Open Policy Studio and Policy Stress Lab.	The same audit is replayed through access-first, balanced, and strict guardrail policies without letting policy override the core gate.	/api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/policy-stress
7. Show non-bypassable counterfactuals	Open Counterfactual Fairness Lab.	The lab identifies which market metrics must genuinely improve and which gates require a fresh audit instead of slider overrides.	/api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/counterfactuals
8. Run adversarial market integrity probes	Open Market Integrity Red Team.	Deterministic probes simulate liquidity withdrawal, spoofed imbalance, volatility cascade, funding squeeze, and reference-price gaps.	/api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/red-team

3.3 Step-by-Step Runbook

Follow these actions in the dashboard and explain the mechanism under each surface.

Step	Action	Underlying mechanism	Proof
9. Open the evidence pack	Open Evidence Pack and copy the JSON bundle.	Every nested report is generated from the same audit hash and linked by proof URLs.	/api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/evidence-pack
10. Anchor the proof	Open On-chain Audit Anchor Kit.	The audit hash can be anchored to make hindsight rewriting much harder while keeping execution and custody out of scope.	/api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/anchor-proof
11. Show the agentic loop	Open Agent Mission Control and Active Agents.	The action queue can refresh, compare, size, hold, review, reset, or paper execute only when the audited gate allows it.	/api/agents/mission?audit_hash=7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b
12. Close with unsafe-market behavior	Switch to Manipulated and show execution locked.	The fallback manipulated scenario deterministically worsens spread, depth, wick, funding, anomaly score, and stop-hit probability.	/api/analysis?symbol=BTCUSDT&category=linear&scenario=manipulated

4. Architecture And Data Flow

The architecture separates data intake, strategy proposal, independent review, route gating, and evidence packaging.

Layer	What happens	Why it matters
Market data	Bybit public data or deterministic fallback snapshots create candles, book levels, ticker fields, funding, and open interest.	The demo remains stable while still supporting live public data.
Feature engineering	Spread, depth, impact, volatility, RSI, momentum, wick pressure, book imbalance, funding, and liquidity score are derived.	The model reasons over explicit market structure, not vague chart vibes.
Agents	Strategy, risk, manipulation, regime, forecast, backtest, planner, memory, narrator, fairness, and route steward checks debate the setup.	The strategy proposer cannot execute by itself.
Verifier layer	Receipt, cohorts, provenance, router, policy stress, counterfactuals, red-team, anchor proof, evidence pack, and readiness runbook are generated from one audit hash.	Judges can verify claims without trusting the UI.

5. Scenario Lab

The project proves selectivity: the same system can approve, observe, or block depending on market quality.

Scenario	Decision	Fairness	Liquidity	Why it matters
Calm	approved / SHORT	89/100	80/100	FairFlow approves a small short paper order with mandatory guardrails.
Volatile	blocked / NO_TRADE	67/100	51/100	FairFlow blocks execution after AI committee review: ML anomaly detector flagged extreme market-structure risk.
Manipulated	blocked / NO_TRADE	50/100	28/100	FairFlow blocks execution after AI committee review: ML regime classifier detected fragile liquidity.

Interpretation

A calm audit may allow guarded paper execution, while volatile and manipulated scenarios demonstrate restraint. No-trade is a product feature because it records why market structure is unhealthy.

6. Fairness Passport And Retail Inclusion

Fairness is scored before execution is discussed.

RECEIPT	COHORTS	HIDDEN COST
85.5/100	inclusive	3.8 bps
BGA alignment	4 pass / 0 watch / 0 block	retail friction

- Information parity: pass - Scenario data is deterministic and reproducible for demos, so judges can replay the same decision.
- Execution parity: pass - Top-of-book costs are low enough that a small retail-sized order is not obviously disadvantaged.
- Manipulation exposure: pass - Manipulation and liquidity-trap signals are inside the safe envelope.
- Retail risk protection: pass - Trade sizing, stop loss, and leverage keep the proposed exposure inside the retail protection envelope.
- Auditability: pass - The report includes a decision trace, agent rationales, safeguards, and a hashable payload.

7. Model And Data Provenance

The project documents what is live, what is derived, and what is only a prototype signal.

Component	Type	Inputs	Limits
ML Market Regime Classifier	ml_signal	rolling returns, volatility, RSI	Nearest-centroid regime signal is lightweight and should not be treated as a production predictive model.
ML Manipulation Analyst	ml_signal	wick size, volume ratio, spread	Robust anomaly scoring can flag stress but cannot prove intent or manipulation.
Uncertainty Forecast Agent	risk_model	recent volatility, momentum, stop distance	Short-horizon distribution is a guardrail, not a guaranteed forecast.
Fairness Passport and Policy Gate	execution_guard	information parity, execution parity, manipulation exposure	Fairness score is a transparent rubric, not a regulatory suitability determination.
Audit Ledger and Anchor Proof	audit_infrastructure	canonical decision payload, receipt metadata, contract arguments	SQLite persistence is local prototype infrastructure.

8. Risk Sizing And Policy Stress

Risk controls are applied before any paper route can proceed.

SIZING	POLICY STRESS	ALLOWED POLICIES
\$2,500 Sizing is executable under the current FairFlow audit gate and retail exposure caps.	fragile greenlight 45.0/100 stability	1/3 access-first, balanced, strict

- Access-first retail review: needs_review - first breaking check: none
- Balanced BGA guardian: compliant - first breaking check: none
- Strict institutional controls: blocked - first breaking check: Fairness floor

9. Fair Execution Router

The router asks whether the execution method itself is fair enough for retail paper routing.

Route	Status	Slippage	Fairness	Reason
Immediate guarded market	available	6.8 bps	76/100	Fastest paper route, but it pays the most visible spread and impact.
Post-only limit	recommended	2.5 bps	89/100	Most retail-friendly when the book is healthy because it avoids crossing the spread.
TWAP micro-slices	available	4.2 bps	82/100	Splits intent across time so larger paper routes do not consume too much top-of-book depth at once.
Maker ladder	available	3.5 bps	86/100	Places smaller passive paper orders at multiple fair prices to reduce impact without chasing fills.
Hold and re-audit	available	0.0 bps	84/100	Best route when fair execution cannot be proven from the current audit.

Router proof

Post-only limit is the fairest paper route for this audit: 89.3/100 route fairness with 2.5 bps expected slippage.

- Expected slippage uses spread, Fairness Passport hidden cost, and 25k impact pressure.
- Information leakage and manipulation exposure are scored separately so a cheap route can still be rejected.
- Recommended routes are paper-only and tied to the immutable audit hash.
- Hold/re-audit is a valid recommendation when route fairness cannot be proven.

10. Counterfactual Fairness Lab

The lab explains what must actually improve before a future audit can be safer.

VERDICT	TOP BLOCKER	UNLOCKABLE
already fair 94.0/100 readiness	No active blocker largest guardrail gap	Yes current audit state

- Fairness score floor: already_clear - current 88.663 target 80.0
- Hidden-cost ceiling: already_clear - current 3.8 target 8.0
- Anomaly-risk ceiling: already_clear - current 30.145 target 45.0
- Liquidity floor: already_clear - current 79.859 target 55.0
- Leverage ceiling: already_clear - current 2.0 target 3.0
- Stop-hit probability ceiling: already_clear - current 0.0 target 45.0
- Core FairFlow execution gate: already_clear - current approved / SHORT / fair_to_execute target approved / actionable / fair_to_execute

11. Market Integrity Red Team

Adversarial probes show where the system halts under worse market structure.

Probe	Status	Trigger	Mitigation
Liquidity rug pull	watch	Hidden-cost cap	Lock execution, shrink notional to zero, and wait for top-depth recovery before re-running the audit.
Spoofed imbalance flip	watch	Anomaly cap	Require anomaly review and avoid market orders when imbalance and wick/volume pressure disagree.
Volatility stop cascade	pass	Hidden-cost cap	Keep paper execution locked unless stop-hit probability and realized volatility both cool down.
Funding squeeze	pass	No guardrail breach	Throttle leverage, require lower notional, and disclose funding/open-interest crowding in the receipt.
Reference-price gap	pass	No guardrail breach	Require fresh public data, anchor the stale audit separately, and refuse execution from old reports.

This is a rare stronger approval: the trade idea survives cost, anomaly, liquidity, and stop-risk perturbations.

12. Evidence Pack And Audit Hash

The evidence pack is the machine-readable verifier artifact for a single audit hash.

VERIFICATION	REPORTS	CLAIMS
93.8/100	10	10
fairflow-evidence-pack-v1	nested proof artifacts	verified, watch, blocked

Claim	Status	Explanation
Audit is replayable by hash	verified	The full decision payload is recoverable through SHA-256 audit hash 7193e89a69c5...
Retail fairness was evaluated	verified	Fairness Passport verdict is fair_to_execute with 88.7/100 and hidden cost 3.8 bps.
Cohort affordability is disclosed	verified	Cohort simulator reports inclusive: 4 pass, 0 watch, 0 block.
No-trade can be a protective outcome	watch	The current audit is not a no-trade, so protective behavior is demonstrated through stress and red-team reports.
Model and data provenance are disclosed	verified	Provenance card documents 3 data sources and 5 AI/risk components.
Governance stress was replayed	watch	Policy Stress Lab verdict is fragile_greenlight with stability score 45.0.
Execution route fairness was checked	verified	Fair Execution Router verdict is route_ready; recommended route is Post-only limit with max paper notional 2950 USDT.
Counterfactual unlock path is bounded	verified	Counterfactual verdict is already_fair; top blocker is No active blocker.
Market integrity probes were run	verified	Red-team verdict is resilient; 0/5 probes trip hard blocks.
On-chain anchor payload is prepared	verified	Anchor proof exposes anchorDecision(bytes32,string,string,string) with payload hash ea331aa38ee8...

13. On-chain Audit Anchor

Blockchain is used as a verification handoff, not as live trading or custody.

Contract: FairFlowAudit in contracts/FairFlowAudit.sol

Function: anchorDecision(bytes32,string,string,string)

Decision hash: 0x7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b

Payload hash: ea331aa38ee850f6ed3449f132c9598e80626075df4df0e2fd629275eb148c57

- Fetch the full audit report from audit_url and confirm the audit_hash matches decisionHash.
- Fetch the fairness receipt from receipt_url and confirm it references the same audit_hash.
- Recompute SHA-256 over the canonical metadata JSON with sorted keys and compact separators.
- Call FairFlowAudit.isAnchored(decisionHash) after transaction submission to verify on-chain presence.
- Read the DecisionAnchored event and compare symbol, action, reporter, timestamp, and metadataURI.

14. Agentic Mission Control

The project is agentic, but bounded by guardrails and paper-only actions.

Agent	Status	Finding
Data Scout	complete	fallback:calm produced a fresh audited report for BTCUSDT.
Regime Analyst	watch	Regime is mean reversion.
Manipulation Sentinel	complete	Anomaly model reads normal at 30/100.
Risk Allocator	complete	Proposed action is SHORT at 2.0x leverage.
Fairness Auditor	complete	FairFlow finds enough parity, protection, and auditability for a capped paper route.
Execution Operator	complete	Route style is limit; side is Sell.
Route Steward	complete	Post-only limit is recommended; router verdict is route ready.
Memory Curator	complete	Session holds 1246 audits, 0 accepted orders, and 0 blocked orders.

Final recommendation: Mission can route a guarded paper SHORT via Post-only limit after sizing and audit review.

15. Watchlist And Impact Ledger

The project can scan a crypto universe and summarize cumulative market-health evidence.

Symbol	Rank score	Fairness	Reason
BTCUSD T	103.2	89/100	Fairness 89/100, liquidity 80/100, anomaly 30/100, gate approved, market-depth adjustment +6.
ETHUSD T	100.6	88/100	Fairness 88/100, liquidity 79/100, anomaly 32/100, gate approved, market-depth adjustment +4.
SOLUSD T	98.3	88/100	Fairness 88/100, liquidity 78/100, anomaly 31/100, gate approved, market-depth adjustment +2.
BNBUSD T	97.3	88/100	Fairness 88/100, liquidity 77/100, anomaly 30/100, gate approved, market-depth adjustment +1.
LINKUSD T	94.2	88/100	Fairness 88/100, liquidity 75/100, anomaly 33/100, gate approved, market-depth adjustment -1.
DOGEUS DT	92.4	93/100	Fairness 93/100, liquidity 100/100, anomaly 42/100, gate approved, market-depth adjustment -9.

50 audits reviewed: 33 approved, 0 observed, 17 blocked, and 17 ended in no trade. Estimated avoided hidden-cost drag is 93.27 USDT.

16. Submission Kit And 2-Minute Video Handoff

This is the final judge-facing package for recording and submitting the project.

RUNTIME	ASSETS	CHECKLIST
120 sec	6	6
6 video segments	PDF, deck, script, endpoint, anchor	submission readiness checks

Time	Segment	Dashboard action
0:00-0:15	Thesis	Open on the Competition Runway with the calm decision, fairness score, hidden cost, and audit hash visible.
0:15-0:34	Problem	Point at hidden cost, liquidity, manipulation risk, and the Fairness Passport.
0:34-0:54	Agentic guardrail	Open Active Agents, Mission Control, and Fair Execution Router.
0:54-1:18	Live demo path	Use Competition Runway: calm approval, submission proof, then manipulated no-trade.
1:18-1:41	No-trade proof	Show Calm approved versus Manipulated blocked; emphasize no-trade as a measurable protection.
1:41-2:00	Why FairFlow wins	Copy the evidence pack and show the audit hash as the final proof handoff.

Required submission assets

- Complete walkthrough PDF: `output/pdf/fairflow_guardian_complete_walkthrough.pdf`
- Two-minute video deck: `output/presentation/fairflow_2_minute_video_deck.pptx`
- Two-minute narration script: `output/presentation/fairflow_2_minute_video_script.txt`
- Verifier evidence endpoint:
`/api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/evidence-pack`

17. Testing, Boundaries, And Honest Limitations

This is where the project earns trust by saying what it does not do.

Validation run by the project:

- Backend API tests cover audit lookup, receipts, cohorts, anchor proof, judge brief, readiness runbook, provenance, counterfactuals, router, red team, evidence pack, scenario comparison, market series, risk sizing, portfolio, mission control, watchlist, and policy evaluation.
- Frontend production build validates TypeScript contracts and the Next.js dashboard bundle.
- Browser checks confirm the readiness and router panels render, buttons work, and mobile layout avoids horizontal overflow.

Known limitations:

- Prototype only; not investment advice and not a regulated suitability engine.
- Live data uses public Bybit endpoints and can fall back to deterministic scenarios.
- ML-style signals are lightweight explainable models intended for guardrails, not production alpha.
- Smart contract integration prepares anchor arguments but does not submit transactions.

18. Final Demo Script

Use this exact closing sequence.

FairFlow Guardian is not trying to win a pure PnL contest. It is an explainable, paper-only trading safety layer that uses public market data, AI-style agents, retail fairness checks, route fairness, adversarial probes, and audit hashes to prove when a trade is fair enough to review and when the healthiest market action is no trade.

Close by switching to Manipulated. Say that this is not a failed trade. This is FairFlow doing its job: refusing execution, recording the reasons, and preserving proof that the decision was made before the outcome.

Proof links to copy during judging:

</api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b>
</api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/hackathon-readiness>
</api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/judge-brief>
</api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/evidence-pack>
</api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/model-provenance>
</api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/execution-router>
</api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/counterfactuals>
</api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/policy-stress>
</api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/red-team>
</api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/receipt>
</api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/retail-cohorts>
</api/audits/7193e89a69c5f1ae682225473cba0f4f64bb58690b08376ca1a8c8a263a3b24b/anchor-proof>
</api/impact?limit=50>